Data Management – Relational Algebra
Queries 1 to 15 (RLMS)

Work distribution

Team member	Queries
Souhayl MOUJAHID	1 to 4
Ahmad KHRISS	5 to 8
Adam NAJI	9 to 12
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Group repository: `https://github.com/NJ-adam/data-Ahmad-Adam-Souyhayl-Ilyas`

Query 1: First and last names of persons who have made at least one equipment reservation

- **Algebra expression:**

$$\pi_{firstName,\,lastName}\Big(\pi_{madeByPid}(Reservation) \bowtie_{madeByPid=pid} Person\Big)$$

- **Difficulty:** the reservation and person keys have different names ($madeByPid$ and pid), so a condition join is needed instead of a natural join.

Query 2: IDs of persons who are either attached to a laboratory or act as PI of at least one research project

- **Algebra expression:**

$$\pi_{pid}(Attached) \;\cup\; \pi_{pid}\Big(\sigma_{role=\text{'PI'}}(Participates)\Big)$$

- **Difficulty:** the two conditions come from different relations, so each side must be projected to the single attribute pid before the union (union-compatible schemas).

Query 3: IDs of laboratories located in the building 'College of Computing' or containing at least one equipment unit whose model belongs to the category 'CPU Server'

- **Algebra expression:**

$$\pi_{labId}\Big(\sigma_{building=\text{'College of Computing'}}(Laboratory)\Big)$$
$$\cup$$
$$\pi_{labId}\Big(\sigma_{category=\text{'CPU Server'}}(EquipmentModel) \bowtie_{modelId=EquipmentUnit.modelId} EquipmentUnit\Big)$$

- **Difficulty:** the category is stored in *EquipmentModel* while the laboratory is stored in *EquipmentUnit*, so a join on *modelId* is needed before the union with the building condition.

Query 4: IDs of laboratories that contain at least one 'CPU Server' unit and at least one 'GPU Server' unit

- **Algebra expression:**

$$\pi_{labId}\Big(\sigma_{category=\text{'CPU Server'}}(EquipmentModel) \bowtie_{modelId=EquipmentUnit.modelId} EquipmentUnit\Big)$$
$$\cap$$
$$\pi_{labId}\Big(\sigma_{category=\text{'GPU Server'}}(EquipmentModel) \bowtie_{modelId=EquipmentUnit.modelId} EquipmentUnit\Big)$$

- **Difficulty:** a single unit cannot have two categories, so one selection with "CPU $\wedge$ GPU" returns nothing. Each category must be computed separately and the results intersected.

Query 5: Faculty members who supervise every laboratory located in the building 'College of Computing'

- **Algebra expression:**

$$\begin{aligned} CCLabs &:= \pi_{labId}\big(\sigma_{building=\text{'College of Computing'}}(Laboratory)\big) \\ SuppCC &:= \pi_{supervisorPid,\, labId}(Laboratory) \;/\; CCLabs \\ Output &:= Faculty \bowtie_{Faculty.pid=SuppCC.supervisorPid} SuppCC \end{aligned}$$

- **Difficulty:** there is no *Supervises* table, since it was absorbed into *Laboratory.supervisorPid*, so the dividend of the division must be built from *Laboratory* itself.

Query 6: IDs of persons who participate in every research project funded by at least one budget that also funds the laboratory with labId $= 2$

- **Algebra expression:**

$$\begin{aligned} B2 &:= \pi_{budgetLine}\big(\sigma_{labId=2}(FundsLab)\big) \\ Target &:= \pi_{projectCode}\big(FundsPrj \bowtie B2\big) \\ Part &:= \pi_{pid,\, projectCode}(Participates) \\ Output &:= Part \;/\; Target \end{aligned}$$

- **Difficulty:** the set of projects to divide by is defined indirectly, through budgets shared with laboratory 2, so it must be computed in two steps (lab $\rightarrow$ budgets $\rightarrow$ projects) before the division.

Query 7: Pairs of persons (p_1, p_2) such that p_1 has made more reservations than p_2

- **Algebra expression:**

$$\begin{aligned} J &:= \pi_{pid,\, resId}\big(Person \bowtie_{pid=madeByPid} Reservation\big) \\ Cnt &:= \gamma_{pid;\ \text{COUNT}(resId)\rightarrow n}(J) \\ Output &:= \pi_{\$1,\,\$3}\big(Cnt \bowtie_{\$2>\$4} Cnt\big) \end{aligned}$$

- **Difficulty:** persons with no reservation must still be counted (with $n = 0$) since they can be a valid p_2, so a left outer join with *Person* is needed and COUNT($resId$) ignores the NULLs.

Query 8: IDs of persons who have reserved equipment units located in at least two different laboratories

- **Algebra expression:**

$$R := \pi_{madeByPid,\, labId}\Big(\pi_{resId,\, madeByPid}(Reservation) \bowtie Reserves \bowtie \pi_{serialNo,\, labId}(EquipmentUnit)\Big)$$

$$Output := \pi_{madeByPid}\Big(\sigma_{k \geq 2}\big(\gamma_{madeByPid;\ \mathrm{COUNT}(labId) \to k}(R)\big)\Big)$$

- **Difficulty:** there is no direct link between a person and a laboratory (the path goes through *Reservation*, *Reserves*, *EquipmentUnit*), and the shared *status* attribute forces projections before the natural joins.

Query 9: IDs of reservations submitted in September 2026 that involve equipment units located in the laboratory named 'AI Movement'

- **Algebra expression:**

$$\begin{aligned}\pi_{resId}\Big(&\sigma_{submissionTS \geq \text{'2026-09-01'} \,\wedge\, submissionTS < \text{'2026-10-01'}}(Reservation)\\ &\bowtie_{resId}\ \pi_{resId,\, serialNo}(Reserves)\\ &\bowtie_{serialNo}\ \pi_{serialNo,\, labId}(EquipmentUnit)\\ &\bowtie_{labId}\ \pi_{labId}\big(\sigma_{name=\text{'AI Movement'}}(Laboratory)\big)\Big)\end{aligned}$$

- **Difficulty:** the laboratory is reached through a chain of four relations, and the month is expressed as a half-open interval ($\geq$ Sept. 1 and $<$ Oct. 1) so that the whole of September is included.

Query 10: IDs of persons who are PI of at least two distinct research projects

- **Algebra expression:**

$$\pi_{pid}\Big(\sigma_{A.pid = B.pid \,\wedge\, A.projectCode \neq B.projectCode}\big(\rho_A(\sigma_{role=\text{'PI'}}(Participates)) \times \rho_B(\sigma_{role=\text{'PI'}}(Participates))\big)\Big)$$

- **Difficulty:** "at least two" cannot be expressed with a single tuple, so the relation must be crossed with a renamed copy of itself and filtered on same *pid* but different *projectCode*.

Query 11: IDs of approvers who approved reservations involving units located in more than one laboratory

- **Algebra expression:**

$$R := \pi_{approverPid,\, labId}\Big(\pi_{serialNo,\, labId}(EquipmentUnit) \bowtie_{serialNo} Reserves \bowtie_{resId} \pi_{resId,\, approverPid}(Reser$$

$$Output := \pi_{A.approverPid}\Big(\sigma_{A.approverPid = B.approverPid \,\wedge\, A.labId \neq B.labId}\big(\rho_A(R) \times \rho_B(R)\big)\Big)$$

- **Difficulty:** the approver is linked to a laboratory only through reservations and units, so R must be built first, then self-joined on same approver but different laboratory.

Query 12: Persons who made no reservation whose planned interval contains November 6, 2026

- **Algebra expression:**

$$\pi_{pid}(Person) - \rho_{pid \leftarrow madeByPid}\Big(\pi_{madeByPid}\big(\sigma_{plannedStart<\text{'2026-11-07'} \wedge plannedEnd\geq\text{'2026-11-06'}}(Reservation)\big)\Big)$$

- **Difficulty:** a negative condition ("no reservation") cannot be expressed with a join or selection, so it is computed as a set difference from all persons, which also keeps persons with no reservation at all.

Query 13: Laboratories whose number of equipment units is below the average number of equipment units per laboratory

- **Algebra expression:**

$$\begin{aligned} T_1 &:= \gamma_{LabID;\ \text{COUNT}(UnitID)\rightarrow numEquip}(EquipmentUnit) \\ T_2 &:= \gamma_{\text{AVG}(numEquip)\rightarrow avgEquip}(T_1) \\ Result &:= \pi_{LabID}\big(\sigma_{numEquip<avgEquip}(T_1 \times T_2)\big) \end{aligned}$$

- **Difficulty:** the average of a count requires two levels of aggregation (count per laboratory, then average of the counts), and the single-row average must be attached to every laboratory with a cross product.

Query 14: For each laboratory, the person who has made the greatest number of distinct approved reservations involving units located in that laboratory

- **Algebra expression:**

$$\begin{aligned} T_1 &:= \sigma_{Status=\text{'Approved'}}(Reservation) \bowtie EquipmentUnit \\ T_2 &:= \gamma_{PersonID,\,LabID;\ \text{COUNT}(ReservationID)\rightarrow numRes}(T_1) \\ T_3 &:= \gamma_{LabID;\ \text{MAX}(numRes)\rightarrow maxRes}(T_2) \\ Result &:= \pi_{PersonID,\,LabID}\big(T_2 \bowtie_{T_2.LabID=T_3.LabID \wedge T_2.numRes=T_3.maxRes} T_3\big) \end{aligned}$$

- **Difficulty:** an aggregate cannot be compared directly with a row, so the maximum per laboratory is computed separately and joined back with the counts, which also keeps ties.

Query 15: IDs of persons who made at least three equipment reservations during the year 2025

- **Algebra expression:**

$$\begin{aligned} T_1 &:= \sigma_{Date\geq\text{'2025-01-01'} \wedge Date\leq\text{'2025-12-31'}}(Reservation) \\ T_2 &:= \gamma_{PersonID;\ \text{COUNT}(ReservationID)\rightarrow numRes}(T_1) \\ Result &:= \pi_{PersonID}\big(\sigma_{numRes\geq 3}(T_2)\big) \end{aligned}$$

- **Difficulty:** "at least three" requires counting, so the year filter must be applied first and the condition $numRes \geq 3$ is selected on the result of the grouping.